

WARRUWI, Australia

WMO: 941390

Lat: 11.6503S

Lon: 133.3797E

Elev: 20

StdP: 101.09

Time Zone: 9.50 (AUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.0	20.9	8.8	7.1	23.8	10.7	8.0	24.6	9.1	25.6	8.4	25.3	4.0	130	0.426

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	5.4	33.7	25.8	33.1	25.8	32.6	25.8	28.5	30.1	27.6	30.4	27.2	30.5	3.9	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
28.3	24.6	29.0	27.0	22.7	29.0	26.2	21.7	29.1	92.7	29.9	88.3	30.5	86.0	30.6	32.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.3	7.5	7.0	18.4	35.1	1.2	0.5	17.6	35.5	16.9	35.8	16.2	36.1	15.4	36.5		
(5)			12.6	28.5	1.0	0.6	11.9	28.9	11.4	29.2	10.8	29.6	10.1	30.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.7	28.8	28.6	28.5	28.4	27.4	25.6	25.1	25.4	27.1	28.7	29.6	29.7
	DBStd	1.94	1.24	1.24	1.10	1.01	1.13	1.55	1.24	1.10	1.08	0.97	0.78	1.34
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6474	584	521	573	551	541	467	468	479	513	580	587	611
	CDD18.3	3432	326	287	315	301	282	217	209	221	263	322	337	353
	CDH23.3	37496	3988	3389	3674	3517	2931	1664	1398	1587	2540	3807	4337	4664
	CDH26.7	13541	1611	1264	1342	1296	850	321	204	272	683	1481	1985	2232
Wind	WSAvg	3.6	3.4	3.3	2.8	3.6	4.6	4.5	4.6	4.2	3.6	3.2	2.7	2.6
Precipitation	PrecAvg	1192	282	256	241	126	20	2	2	0	2	10	68	195
	PrecMax	1790	583	502	547	652	185	46	44	6	16	67	458	425
	PrecMin	609	58	34	38	2	0	0	0	0	0	0	0	39
	PrecStd	312	117	121	139	135	32	7	7	1	3	16	89	109
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.7	33.3	33.4	33.1	32.0	31.0	30.2	30.5	32.0	33.5	34.0	34.7
		MCWB	26.7	26.6	26.8	25.6	24.7	23.4	21.7	21.2	23.1	23.8	25.2	25.8
	2%	DB	32.9	32.5	32.3	32.1	31.2	30.0	29.2	29.6	31.1	32.6	33.3	33.9
		MCWB	26.6	26.6	26.5	25.3	24.4	23.1	21.6	21.1	23.0	24.5	25.3	26.0
	5%	DB	32.2	31.8	31.7	31.6	30.6	29.2	28.6	28.9	30.4	32.0	32.8	33.2
		MCWB	26.6	26.5	26.4	25.2	24.2	22.7	21.5	21.1	23.0	24.6	25.3	26.0
	10%	DB	31.5	31.2	31.0	31.0	30.0	28.5	27.9	28.2	29.7	31.4	32.2	32.6
		MCWB	26.5	26.3	26.3	25.0	23.9	22.3	21.3	21.0	22.9	24.5	25.2	26.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	29.0	27.8	30.8	27.0	26.7	26.9	26.6	24.3	25.7	29.3	27.1	28.5
		MCDB	30.7	31.7	31.1	30.9	29.4	27.0	26.8	27.8	29.7	29.3	31.7	30.3
	2%	WB	28.0	27.3	29.0	26.6	26.0	24.6	24.2	23.7	25.0	26.2	26.6	27.6
		MCDB	30.5	31.0	29.3	30.2	29.5	27.8	26.1	27.1	28.7	30.3	31.2	30.8
	5%	WB	27.3	27.0	27.5	26.2	25.5	24.2	23.4	23.2	24.5	25.7	26.2	27.1
		MCDB	30.5	30.6	30.0	29.7	29.0	27.6	26.3	26.6	28.3	30.1	30.8	31.0
	10%	WB	26.9	26.7	26.9	25.9	25.1	23.7	22.8	22.8	24.1	25.3	25.9	26.7
		MCDB	30.4	30.1	29.9	29.3	28.6	27.3	26.2	26.3	28.0	29.7	30.5	30.7
Mean Daily Temperature Range	5% DB	MDBR	5.0	4.8	5.1	5.4	5.5	6.1	6.1	6.3	6.0	5.8	5.9	5.4
		MCDBR	5.9	5.6	5.9	5.9	5.8	6.1	6.3	6.7	6.5	6.4	6.4	6.1
		MCWBR	2.2	2.2	2.4	2.1	2.5	3.2	3.4	3.4	2.5	2.2	1.8	2.0
	5% WB	MCDBR	5.3	5.2	5.3	5.2	5.1	5.5	5.6	5.6	5.6	5.7	5.9	5.5
		MCWBR	2.6	2.3	3.1	2.1	2.4	2.8	3.2	2.6	2.2	2.0	2.0	2.3
Clear-Sky Solar Irradiance	taub	0.415	0.407	0.396	0.379	0.372	0.361	0.354	0.365	0.409	0.427	0.426	0.413	
	taud	2.520	2.538	2.572	2.608	2.598	2.583	2.594	2.541	2.411	2.379	2.425	2.507	
	Ebn at Noon	928	930	923	908	884	878	893	909	897	901	912	928	
	Edh at Noon	113	110	104	96	91	90	91	101	121	128	123	114	
All-Sky Solar Radiation	RadAvg	5.22	5.42	5.30	5.46	5.24	5.04	5.30	6.00	6.62	6.89	6.69	5.73	
	RadStd	0.64	0.74	0.62	0.63	0.35	0.18	0.17	0.17	0.22	0.27	0.26	0.60	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)	+0.40	N/A	N/A	N/A	N/A	-0.47	N/A	N/A	N/A	N/A

Nomenclature: See separate page